mixcuref Manual

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1 Overview

`mixcuref` fits Weibull mixture cure models by maximum likelihood or Firth-type penalized likelihood. The package provides direct estimation, profile-likelihood confidence intervals, nested likelihood-ratio tests, and utilities for separation diagnostics.

The main user-facing functions are:

- `mixcure.penal.est()` for model fitting
- `mixcure.penal.profile.CI()` for profile-likelihood confidence intervals
- `mixcure.penal.1d.nested.lrt()` and `mixcure.penal.2d.nested.lrt()` for nested tests
- `separation.detection()` for sparse-data diagnostics
- `fit_mixture_cure()` as a thin helper used by the meta-layer
- `mixcuref_fit()` as the adapter used by `Mixcuref.Meta`

2 Package data

The package includes three real breast cancer datasets:

- `ANNbcBMdat1`: high event rate, univariate example
- `ANNbcBMdat2`: lower event rate, univariate example
- `ANNbcBMdat5`: lower event rate, multivariable example

3 Fitting a mixture cure model

A mixture cure model is specified using a standard survival formula `Surv(time, status) ~ covariates`. Covariates are used in both the cure and latency components.

```
library(mixcuref)
library(survival)

data(ANNbcBMdat1)

fit_mle <- mixcure.penal.est(
  Surv(Time, CENS == 1) ~ Her2,
  data = ANNbcBMdat1,
  init = c(1, -0.1, -10, 1, 1),
  pl = FALSE
)

fit_ftpl <- mixcure.penal.est(
  Surv(Time, CENS == 1) ~ Her2,
  data = ANNbcBMdat1,
  init = c(1, -0.1, -10, 1, 1),
  pl = TRUE
)

fit_mle$coefficients
fit_ftpl$coefficients
```

4 Multivariable example

```
data(ANNbcBMdat5)

fit5 <- mixcure.penal.est(
  Surv(Time, CENS == 1) ~ Her2 + LuminalA + TN + MENS0 + TUMCT,
  data = ANNbcBMdat5,
  init = c(5, rep(0, 5), -10, rep(0, 5), 0.1),
  pl = TRUE
)

fit5$coefficients
coef(fit5)
vcov(fit5)
```

5 Profile-likelihood confidence intervals

```
plci1 <- mixcure.penal.profile.CI(
  Surv(Time, CENS == 1) ~ Her2,
  data = ANNbcBMdat1,
  init = c(1, -0.1, -10, 1, 1),
  pl = FALSE,
  apct = 0.05
)

plci1$coefficients
```

6 Nested likelihood-ratio tests

```
lrt1 <- mixcure.penal.1d.nested.lrt(
  Surv(Time, CENS == 1) ~ Her2,
  data = ANNbcBMdat1,
  init = c(1, -0.1, -10, 1, 1),
  pl = FALSE,
  loglik = fit_mle$loglikelihood
)

lrt1
```

7 Separation diagnostics

```
sep_diag <- separation.detection(fit5, pl = TRUE, nsteps = 20)
sep_diag
```

8 Meta-layer adapters

`mixcuref` can still be used directly, but it also provides helper functions that support the `Mixcuref.Meta` package.

```
fit_from_helper <- fit_mixture_cure(
  incidence = ~ Her2,
  latency = ~ Her2,
  time = ANNbcBMdat1$Time,
  status = ANNbcBMdat1$CENS,
  data = ANNbcBMdat1,
  control = list(pl = FALSE)
)

fit_from_helper
```

```
# spec is a model specification object created by Mixcuref.Meta
# mixcuref_fit(spec, control = list(pl = FALSE))
```

9 Notes

- `pl = FALSE` requests standard maximum likelihood.
- `pl = TRUE` requests Firth-type penalized likelihood.
- Initial values must be supplied in the order: cure coefficients, survival coefficients, Weibull shape parameter.
- For categorical covariates with more than two categories, create dummy variables before fitting.

10 Session information

Table 1: Package Versions

Package	Version	Built	Remote
survival	3.8.3	R 4.5.0; x86_64-w64-mingw32; 2025-04-11 10:21:40 UTC; windows	
abind	1.4.8	R 4.5.0; ; 2025-04-08 00:42:55 UTC; windows	
R.utils	2.13.0	R 4.5.0; ; 2025-06-16 00:32:17 UTC; windows	